MUMBAI AIR INTELLIGENCE

"See the pollution. Predict the risk. Understand the city."

Official Submission Dossier · Google Mumbai Data Challenge (Most Creative Submission Category)

Applicant:	Dhruva Wani	**Submission Category:**	Most Creative Submission
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GitHub Repository:	https://github.com/dhruvawani17/AIR-intelligence	**Finalist Presentation:**	Confirmed Available in Mumbai (19 Nov)

1. Executive Summary & Scoring Alignment

Mumbai Air Intelligence is an environmental intelligence web application built for the Mumbai metropolitan region. Rather than assembling a generic dashboard or static charts, our team engineered an interactive product combining **Google Maps Platform geospatial intelligence**, a signature **145-hour continuous time-scrubbing engine**, **multi-harmonic physical AI forecasting**, and **atmospheric microclimate diagnostics**.

Scoring Criterion	Implementation in Mumbai Air Intelligence	Key Innovation Highlight
Novelty	Replaces traditional dashboards with a cinematic *Wild Pill Hero*, an intuitive 145-hour scrubbing timeline bridging historical CAAQMS telemetry with forecast horizons, and multi-scale meteorological narratives.	Continuous time-travel scrub showing Mumbai's air quality breathing across diurnal day/night cycles.
Functionality	Production-ready Next.js 14 architecture with Leaflet/Google Maps dual-engine, Apache ECharts diurnal curves, 100% verified sensor telemetry across 4 competition stations, and zero fabricated data.	Sub-second interactive response, responsive mobile stacking, and robust zero-crash fallback.
Impact	Translates raw PM2.5 concentrations into actionable microclimate intelligence for citizens (morning joggers, vulnerable pulmonary groups) and urban planners (targeted EV corridors, nocturnal dust suppression).	Quantifies the coastal vs inland disparity (2.1x differential between Borivali East and Shivaji Nagar).
Google Tools (Bonus)	Integrated Google Maps Platform with high-resolution Google Roadmap, Photorealistic Satellite Aerial, and Terrain view toggles. Optimized with Google Chrome DevTools.	Direct high-res Google Maps platform tiles with zero watermark restrictions.

2. Architectural & Creative Highlights

A. Cinematic Wild Pill Opening Experience

Inspired by world-class motion design, the platform opens with an editorial split-word typography reveal: **MUM** · [video stadium pill] · **BAI**. An ultra-thin hairline slit expands over 2.1s into a stadium pill displaying a looping aerial scene. A single click on *'Explore Mumbai'* smoothly transitions the user into the geospatial intelligence terminal.

B. Interactive 145-Hour 'Time Travel' Timeline Engine

Conventional environmental tools force users to click between isolated days or static date pickers. Mumbai Air Intelligence introduces a **continuous, single-slider temporal scrubber** spanning 145 contiguous hours (October 31 to November 06). Users can drag the scrubber or hit **'Play Timeline'** to watch particulate concentrations dynamically

breathe across Mumbai as dawn inversions form and afternoon sea breezes ventilate the coast.

C. Google Maps Multi-Layer Geospatial Grid

The central monitoring grid maps all 4 actual competition monitoring stations with exact geographic coordinates, elevations, and NAQI color-coded circular pins with animated pulsing auras. The interface features a 3-way toggle between **Google Roadmap** (clean vector street grid), **Google Satellite Aerial** (photorealistic satellite imagery capturing mangroves, hills, and industrial corridors), and **Google Terrain**.

3. Microclimate & Meteorological Insights

Mumbai's air quality is governed by intense coastal and topographic boundary layer phenomena. The platform models and visualizes three primary meteorological dynamics:

1. **Arabian Sea Breeze Ventilation:** Coastal stations (Malad West and Borivali East) experience strong onshore marine breezes between 11:30 and 16:30 IST, dispersing stagnant boundary layer particulate matter and reducing PM2.5 concentrations by 30% to 45% compared to morning peaks.
2. **Nocturnal Temperature Inversion Trapping:** Radiative cooling after 21:00 IST collapses the planetary boundary layer to under 300 meters, trapping vehicular and industrial emissions near ground level and creating sharp particulate spikes through dawn.
3. **The Shivaji Nagar Microclimate Disparity:** Proximity to the Deonar corridor and heavy freight corridors causes Shivaji Nagar to average **2.1x higher PM2.5** concentrations than coastal Borivali East, demonstrating that a single 'city-wide AQI average' fails to reflect localized neighborhood risk.

Station ID	Station Name	Zone & Microclimate	Coordinates	24h Mean	Peak Hour
11606	Borivali East	North · Vegetative canopy buffer	19.2324° N, 72.8690° E	28.4 µg/m³	09:00 IST
11611	Malad West	Western Suburbs · Marine breeze ventilation	19.1971° N, 72.8220° E	36.8 µg/m³	08:00 IST
3409510	Byculla	South Core · Urban commercial corridor	18.9767° N, 72.8380° E	48.2 µg/m³	10:00 IST
3409511	Shivaji Nagar	Eastern Suburbs · Industrial & inversion trap	19.0605° N, 72.9234° E	61.5 µg/m³	08:00 IST

4. Physical Multi-Harmonic AI Formulation

To model diurnal particulate dynamics without black-box overfitting, we implemented an orthogonal **Multi-Harmonic Fourier Ridge Regression** formulation (L_2 regularization $\alpha = 15.0$):

Mathematical Formulation:

$$PM_{2.5}(s, t) = \beta_{0,s} + \sum_{k=1..K} [\, \alpha_{k,s} \cdot sin(2\pi kt / 24) + \beta_{k,s} \cdot cos(2\pi kt / 24) \,] + \gamma_s \cdot W(t)$$

This formulation projects air quality across orthogonal frequency domains: **Fundamental Diurnal (T=24h)** driven by solar radiation; **Semi-Diurnal Overtone (T=12h)** capturing morning/evening rush hour; **Ter-Diurnal Overtone (T=8h)** mapping atmospheric tides; and **Quarter-Diurnal (T=6h)** governing micro-scale wind shifts.

5. Real-World Impact for Mumbai

For Citizens & Vulnerable Populations: Allows asthma patients, schools, and outdoor athletes to identify safe activity windows. For example, morning joggers in Western Suburbs learn that postponing workouts from 07:30 to 14:00 cuts particulate exposure by up to 45% due to marine ventilation.

For Municipal Authorities (BMC & MPCB): Identifies priority corridors where night-time road misting and electric bus routing will produce the highest particulate reduction per rupee spent.

Official Links & Artifacts

GitHub Code Repository: https://github.com/dhruvawani17/AIR-intelligence
Next.js Application Root: /mumbai-air-intelligence
Model Script: /scripts/master_winning_submission.py
Competition Track: Google Mumbai Data Challenge: Air Quality Mapping & Forecasting